

Unit 8 | Chapter 14: The "Brain" (Databases & SQL)

Learning Targets:	
❖ Technical: I can differentiate between Relational (SQL) and Non-Relational (NoSQL) databases and choose the correct structure for a given dataset. (NV 5.4.1) ❖ Technical: I can explain the CRUD lifecycle (Create, Read, Update, Delete) and its application in data management. (NV 5.4.1) ❖ Technical: I can write basic SQL queries using commands such as SELECT, FROM, WHERE, INSERT, and ORDER BY. (NV 5.4.1) ❖ Technical: I can define primary keys and relationships between tables to ensure data integrity. (NV 5.4.1) ❖ Employability: I can demonstrate analytical reasoning and cognitive flexibility to design efficient database schemas that minimize data redundancy. (WRS 1.2.3) ❖ Employability: I can demonstrate probity and professional decorum by identifying the ethical implications of data storage and privacy laws (e.g., GDPR/CCPA). (WRS 1.1.2, 1.1.4)	
Compelling Question:	
How does a "Digital Brain" store, organize, and retrieve millions of pieces of information instantly, and what is the developer's ethical responsibility toward that data?	
Score	Performance Indicators
4.0	I can: ☐ Architect a complex Relational Schema involving multiple tables and foreign keys to eliminate data redundancy, demonstrating advanced analytical reasoning. (NV 5.4.1, WRS 1.2.3) ☐ Write advanced SQL queries involving JOIN operations to retrieve related data from separate tables with systematic resolution. (NV 5.4.1, WRS 1.2.3) ☐ Evaluate a database design for security vulnerabilities (like SQL Injection) and propose solutions, demonstrating strict probity and ethical adherence. (WRS 1.1.2)
3.5	In addition to score 3.0 performance, partial success at score 4.0 content.
3.0	I can: ☐ Successfully execute the full CRUD lifecycle using SQL syntax or a visual database manager. (NV 5.4.1) ☐ Use WHERE clauses and ORDER BY to filter and sort specific datasets based on user requirements, applying analytical reasoning. (NV 5.4.1, WRS 1.2.3) 

	☐ Define a Primary Key for a table and explain why it must be unique to maintain data integrity. (NV 5.4.1) ☐ Differentiate between SQL and NoSQL based on data structure (Tables vs. Documents). ☐ Demonstrate intrinsic motivation and punctuality by completing daily database labs on schedule. (WRS 1.1.1)
2.5	No major errors or omissions regarding score 2.0 content, and partial success at score 3.0 content.
2.0	I can ☐ Define common terminology and identify syntax: ☐ Database, SQL, NoSQL, CRUD, Table, Row (Record), Column (Field), Primary Key, Schema, SELECT, FROM, WHERE, INSERT. (NV 5.4.1) ☐ Identify the importance of punctuality (WRS 1.1.1), probity in data privacy (WRS 1.1.2), professional decorum (WRS 1.1.4), inclusive collaboration (WRS 1.1.5), and applying analytical reasoning to data organization (WRS 1.2.3).
1.5	Partial success at score 2.0 content, and major errors or omissions at 3.0.
1.0	With help, partial success at score 2.0 and score 3.0 content.
0.5	With help, partial success at score 2.0 but not at score 3.0 content.
0.0	Even with help, no success or attempt made.